AREL URBANOZO

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EDUCATION

University of Alabama in Huntsville — Huntsville, AL
Bachelor of Science in Aerospace Engineering — *Expected May 2027*

- GPA: 3.85/4.00 · President's List (F23, S24) · Dean's List (F24, S25, F25) · UAH Merit Scholarship
- Relevant coursework: Mission Design (capstone), Astrodynamics, Systems & Controls, Measurement & Instrumentation, Rocket Propulsion (graduate), Aircraft Stability & Control (graduate)

ENGINEERING EXPERIENCE

UAH IDEAS Lab — ChargerSat-2 — Huntsville, AL
Lead Systems Engineer & Communications Lead (Research Assistant) — *Feb 2026 – Present*

- Lead systems engineering for a 12U NASA Class D CubeSat (on-orbit autonomous-AI demo), coordinating interfaces, requirements decomposition, and verification-matrix closure across 8 subsystem teams through the Phase A MDR/SRR gate.
- Authored the communications plan and link-margin analysis; closed a 50 Mbps X-band LEO downlink at +4.04 dB and showed the 100 Mbps option (+2.03 dB) fails the 3 dB requirement.
- Identified that the baselined X-band radio cannot reach the 8400–8450 MHz DSN deep-space band, driving an S-band cislunar re-plan and FCC Part 5 Experimental licensing as the primary regulatory path.
- Authored the Universal Payload ICD (≈6.5U, ≤8.5 kg, ≤42 W orbit-average); built an RF Mission Workbench (Python MILP scheduler, Three.js globe) for downlink planning.

UAH IDEAS Lab — INTREPID (NASA-funded lunar CubeSat outreach) — Huntsville, AL
Research Assistant — Curriculum, Hardware & Test — *Feb 2026 – Present*

- Authored 17-week Sensor and Testing Team curricula (guides, teacher notes, decks) for high-school CubeSat teams.
- Designed the student payload PCB in KiCad (95×95 mm, 2-layer; RP2040, MPU-6050 IMU, thermistor) from schematic to fab-ready Gerbers in one week.
- Led a deployer velocity test campaign (n = 7, 997 Hz IMU + photogate); proved the IMU captured only 73% of the true 1.08 m/s separation and traced the error to fixture impact (pulse 5.7× narrower than spring theory), ruling out mount resonance.
- Deliver on-site instruction and mentorship at partner high schools; coordinate logistics and test days.

UAH College of Engineering — Huntsville, AL
Teaching Assistant — EGR 101 Intro to Computing for Engineers — *Aug 2025 – May 2026*

- Supported a 20-student section in lecture and office hours, debugging Python/MATLAB; graded ~40 assignments weekly.

UAH Enhanced Teaching & Learning Center — Huntsville, AL
Student Specialist — *Oct 2024 – May 2025*

- Diagnosed A/V failures across 130 technology-enabled classrooms; installed projectors, cameras, and rack hardware.

LEADERSHIP

UAH Space Hardware Club — CanSat 2024, Team 2078 "Shockwave" — Huntsville, AL
Team Lead — *Oct 2023 – Jun 2024*

- Led a 10-person team as a freshman through PDR, CDR, environmental test, flight, and post-flight review; placed 2nd internationally and top U.S. team (score 99.52) at the AAS CanSat Competition.
- Ran component and design trade studies; designed and iterated the flight PCB seven times in KiCad (Teensy 4.1, BMP390, BNO055, XBee telemetry, 3 regulators).

PROJECTS

- **Hooke's Law & CubeSat Deployment Simulator** *(React, TypeScript, Vite, Aug 2026):* Offline physics lab for Chromebooks: Verlet-integrated CubeSat deployer challenge (1U–6U, 1.2–1.8 m/s, ≤15 g), WCAG 2.1 AA.
- **PMI Dynamometer** *(KiCad, C++, MATLAB, Apr 2026):* Torque/RPM dynamometer with custom PCB (Pico 2, HX711 load cell, Hall-effect RPM), MATLAB GUI, and load-cell uncertainty analysis.

SKILLS

Systems Engineering: NASA SE lifecycle, requirements & VCRM, ICDs, trade studies, risk & config. management, design reviews
RF & Mission: Link budgets, TT&C, ground-contact planning, spectrum licensing, power & duty-cycle budgets, astrodynamics
Hardware & Test: PCB design (KiCad, 5+ boards), embedded sensors, TH/SMD soldering, instrumentation & DAQ, test & analysis
Software: Python (NumPy, SciPy), MATLAB, C/C++, CircuitPython, TypeScript/React, Svelte, Git, Inventor, Onshape, LTspice